The Complete Map of AI Audio Models

Categories, Models, Capabilities, Costs, and Licensing

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Before you read

I wrote this report because the audio AI market has become almost impossible to navigate from the outside. Vendor pages compare their own best model to everyone else's worst, benchmark tables measure English read speech and call it "multilingual," and license terms are buried three clicks away from the download button. What follows is my attempt to put the whole field on one page: eight functional categories, the models that matter in each, what they actually cost, and where the legal tripwires are.

A few notes on method. The comparison is organized by *function* rather than architecture, because that is how build-versus-buy decisions actually get made. Prices are list prices as of November 2026, converted to US dollars, and they move constantly — treat every number here as a starting point for a quote, not as a quote. Where I could verify a figure against a vendor page I did; where I could not, the number is an estimate and should be checked before it goes into a budget. Model versions churn monthly, so a name in this report may already have a successor by the time you read it.

I have tried to be honest about the parts that are genuinely uncertain: latency figures depend on region and payload, WER depends on domain, and "supports 99 languages" usually means "supports 99 languages badly, except English." Where the data is thin, I say so.

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1. The short version

1. There are **eight functional categories**: text-to-speech (TTS), speech-to-text (ASR), voice conversion and cloning, music generation, sound-effect generation, singing synthesis, audio understanding, and audio processing/restoration.
2. The real dividing line is **cloud versus open weights**. Cloud gives you quality, uptime, and someone else's on-call rotation; open weights give you control and a fixed GPU bill, at the cost of maintenance and uneven quality.
3. TTS has stratified into three tiers: high-expression models (ElevenLabs v3, Hume Octave), low-latency agent models (Cartesia Sonic, Deepgram Aura-2, ElevenLabs Flash), and small open models (Kokoro, Piper, Chatterbox, CosyVoice).
4. Whisper remains the default "good and cheap" ASR. On English accuracy it has been overtaken by Parakeet/Canary, Nova-3, and AssemblyAI — but for Persian, Whisper plus Azure or Google is still the realistic stack.
5. Music splits into consumer products (Suno, Udio, Riffusion) and open models (ACE-Step, Stable Audio Open, MusicGen, YuE). Full-song vocals remain a commercial advantage.

6. Sound effects grew fastest: text-to-SFX is now routine, and video-to-Foley (MMAudio, HunyuanVideo-Foley) is close behind.
7. Audio understanding has moved to "Omni" models — GPT-4o/5 audio, Gemini, Qwen3-Omni, Kimi-Audio. These are no longer transcription tools; they are voice assistants with ears.
8. Cost should be measured per unit of work: TTS per million characters, ASR per minute, music per generation, self-hosting per GPU-hour. Rule of thumb: below roughly 500 hours of audio per month, APIs win; above it, dedicated GPUs do.
9. Licensing is the main trap in open weights: CC-BY-NC, community licenses, and "open code, closed weights." Always check the license on the *weights*, not the GitHub repo.
10. Legal risk is the most serious issue in the field: voice cloning without consent, and ownership of generated music. For a commercial product, the safe path is a model with an explicit commercial license plus a supplier contract.

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2. The map

#	Category	Input → output	What success means	Typical use
1	**TTS**	text → speech	naturalness, expression, latency, cloning	audiobooks, assistants, ads
2	**ASR**	speech → text	WER, diarization, streaming, language coverage	subtitles, calls, analytics
3	**Voice conversion**	audio + target → new voice	timbre similarity, content preservation, real-time	dubbing, games, creator tools
4	**Music generation**	text/melody → music	musicality, structure, vocals	songs, background, games
5	**Sound effects / Foley**	text/video → audio	sync with picture, realism	film, games, podcasts
6	**Singing synthesis**	lyrics + melody → singing	pronunciation, vibrato, emotion	song production, demos
7	**Audio understanding**	audio → meaning	reasoning, tagging, answering	voice assistants, monitoring
8	**Processing / restoration**	audio → audio	SNR, separation, naturalness	cleanup, stems, mastering

Two secondary axes are useful. By **signal domain**: speech, music, and general/environmental sound. By **model direction**: generative (TTS, music, SFX) versus analytical (ASR, tagging, speaker recognition). Omni models do both.

3. How I compared them

Every model was assessed against the same twelve criteria, so the tables are at least internally consistent.

Criterion	What it measures	Why it matters
Perceptual quality	MOS or arena ranking (TTS Arena, Artificial Analysis)	closest proxy for "does it sound real"
Accuracy (ASR)	WER on clean, noisy, and accented sets	the main purchase decision
Latency	time to first audio, or real-time factor	decides whether an agent feels alive
Language coverage	number of languages, and quality per language	Persian and multilingual work
Cloning	zero-shot, few-shot, fine-tune	personalization
Control	emotion, speed, prosody, pronunciation, audio tags	production quality
Output length	seconds or minutes per request	audiobooks, podcasts
Streaming	WebSocket / real-time support	telephony and agents
Hardware	CPU/GPU, memory, RTF	self-host cost
License	Apache/MIT/CC-BY-NC/community/proprietary	commercial use
Price	per character, minute, or generation	budget
Maturity	docs, SDKs, SLA, stability	operational risk

4. Text-to-speech

4.1 The state of the market

Three generations of TTS are running in production at the same time:

- **Generation 1 (concatenative and parametric):** Tacotron, FastSpeech, VITS. Fast, small, medium quality, still common in embedded products and low-resource languages.
- **Generation 2 (diffusion and LLM-based):** XTTS, F5-TTS, CosyVoice, Chatterbox, Orpheus, Dia. Zero-shot cloning and high expressiveness.
- **Generation 3 (full duplex):** CSM, Moshi, GPT-4o audio, Gemini. TTS as one component of a conversational model that understands context.

4.2 Commercial services

Model / service	What stands out	Latency	Languages	Cloning	Approx. price
ElevenLabs v3	Most expressive model on the market; emotion tags (`[whispers]` , `[laughs]`); 70+ languages	0.5–1 s (non-streaming)	70+	Instant (1 min sample) + professional	$0.06–0.24 per 1k chars; plans $5–$1,320/mo
ElevenLabs Flash v2.5	Same engine, tuned for speed	~75 ms	32+	Limited	Cheaper than v3
ElevenLabs Turbo v2.5	Speed/quality balance	~250–300 ms	32+	Yes	Mid
OpenAI gpt-4o-mini-tts	Instruction-following ("calm, serious tone"); native to the API	~0.5 s	~50 (variable quality)	No	~$0.015/min of audio
OpenAI tts-1 / tts-1-hd	Simple, stable, 13 voices	Fast	~50	No	$15 / $30 per 1M chars
Google Chirp 3 HD	Natural voices, broad coverage, GCP	~0.3 s	50+	Custom Voice (enterprise)	~$30 per 1M chars
Google Gemini TTS	Natural-language control, multi-speaker	Medium	24+	No	Audio-token based (preview)
Azure Neural TTS	Official fa-IR support; full	~0.2 s	140+	Custom Neural	~$15 per 1M (HD: $30)

Model / service	What stands out	Latency	Languages	Cloning	Approx. price
	SSML; streaming			Voice	
Amazon Polly	Cheap, stable, generative voices	~0.3 s	40+	Brand Voice	$4 standard / $16 neural / $30 generative
Cartesia Sonic-3	Very fast, very cheap for conversation; good quality	~40 ms	40+	Instant clone	~$20–40 per 1M chars
Deepgram Aura-2	Built for agents; low latency; SLA	~200 ms	English-first	No	~$30 per 1M chars
Hume Octave	Emotional "acting" controlled by description	Medium	10+	Yes	Per character, above average
PlayHT Play 3.0	Fast cloning, good language coverage	~0.3 s	30+	Yes	From ~$39/mo; API per char
LMNT	Extremely fast, minimal	~50 ms	10+	Yes	~$0.03 per 1k chars
Rime	Character-driven, conversational voices	~0.2 s	10+	Yes	Per character
MiniMax Speech-2.6	High quality at aggressive	Medium	40+	Yes	Low, competitive

Model / service	What stands out	Latency	Languages	Cloning	Approx. price
	prices; 40+ languages				with ElevenLabs
Fish Audio S1	Fast cloning, low price, open ecosystem	Medium	13+	Yes	~$15 per 1M chars
Speechify / Murf / WellSaid / Resemble	Product-first (studio, teams, brand voice)	Medium	20–50	Yes	$20–100/mo subscriptions

> **A note on price.** Measure TTS by the cost of one hour of audio, not by the headline per-character rate. One hour of speech is roughly 90,000 characters. So $15 per 1M characters is about **$1.35 per audio hour**; $30 per 1M is **$2.70**; ElevenLabs on a low-tier plan can reach **$10–20 per hour**.

4.3 Open-weight models

Model	Params	Languages	Cloning	License	Hardware	Strength
Kokoro-82M	82M	8 (en, ja, zh, es, fr, hi, it, pt)	No (voicepacks)	Apache 2.0	Runs on CPU	Unbeatabl quality pe cost; very
Piper	10–60M	30+ (incl. fa_IR)	No	MIT	Raspberry Pi	Light, offl multilingu
XTTS-v2 (Coqui)	~467M	17	6-second sample	CPML (non-commercial)	4–6 GB VRAM	Mature multilingu cloning
F5-TTS	~336M	en, zh (+finetunes)	Zero-shot	Code MIT / weights CC-BY-NC	6–8 GB	High qual trainable
Chatterbox (Resemble)	0.5B	en + 23 (multilingual release)	Yes	MIT	6–8 GB	Adjustable emotion, watermar
Dia-1.6B (Nari)	1.6B	en	Yes	Apache 2.0	10 GB+	Multi-spea dialogue, laughs
Orpheus-3B (Canopy)	3B	en + language variants	Yes	Apache 2.0	12 GB+	Emotion t ~200 ms streaming
Sesame CSM-1B	1B	en	Yes	Apache 2.0	8 GB	Conversat context, natural

Model	Params	Languages	Cloning	License	Hardware	Strength
VibeVoice (Microsoft)	1.5B / 7B	en, zh	Yes	MIT	8–16 GB	Up to 4 speakers, to 90 min
Higgs Audio v2 (Boson)	3B	multilingual	Yes	Apache (code) / open weights	12 GB+	High qual music and singing
IndexTTS-2 (Bilibili)	~1B	zh, en	Zero-shot	Apache 2.0	8 GB	Emotion a duration control
CosyVoice 2/3 (Alibaba)	0.5B	zh, en, ja, ko + dialects	Yes	Apache 2.0	4–6 GB	~150 ms streaming productio quality
Fish Speech / OpenAudio	~0.5B	13	Yes	Code Apache / weights CC-BY-NC-SA	6 GB	Multilingu fast clonir
MegaTTS3	0.45B	zh, en	Yes	Apache 2.0	6 GB	Good clor light
Zonos (Zyphra)	1.6B	en + 7	Yes	Apache 2.0	8 GB	Emotion control, g timbre
Bark (Suno)	~1B	13+	Yes	MIT	6 GB	Non-verba sounds, m

Model	Params	Languages	Cloning	License	Hardware	Strength
Parler-TTS	0.9B	en	No	Apache 2.0	6 GB	Text-descr voice cont
Spark-TTS	0.5B	zh, en	Yes	Apache 2.0 (code)	4 GB	Light, controllab
MOSS-TTSD / FireRedTTS-2	1–3B	zh, en	Yes	Apache 2.0	8–12 GB	Long dialo podcasts

Bottom line on TTS. Best absolute quality: ElevenLabs v3, Hume Octave, MiniMax in commercial; Higgs Audio v2, IndexTTS-2, Chatterbox in open. Best quality per dollar in open: Kokoro (no cloning) and CosyVoice (with cloning). Best for real-time agents: Cartesia, Deepgram Aura-2, ElevenLabs Flash; on the open side, CosyVoice-2 and Orpheus. Best for Persian: Azure Neural/Custom Neural, Google Chirp 3, ElevenLabs (test it), Polly; open options are Piper fa_IR and community VITS/Coqui models trained on ManaTTS.

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5. Speech-to-text

5.1 Commercial

Service / model	Approx. price	WER (clean English)	Streaming	Diarization	Lan
OpenAI whisper-1	$0.006/min	~8–12%	No	No	99
OpenAI gpt-4o-transcribe	~$0.006/min	~5–8%	Limited	No	50+
OpenAI gpt-4o-mini-transcribe	~$0.003/min	~7–10%	Limited	No	50+
Deepgram Nova-3	~$0.0043–0.0077/min	~5–7%	Yes (strong)	Yes	30+
AssemblyAI Universal-3	~$0.0025–0.0045/min	~5–7%	Yes	Yes	99
Speechmatics	~$0.0017–0.0036/min	~6–8%	Yes	Yes	50+
Soniox	~$0.001–0.002/min	~6–9%	Yes	Yes	60+
Gladia	~$0.0025/min	~7–9%	Yes	Yes	100
Rev AI	$0.02/min (AI)	~6–8%	No	Yes	30+

Service / model	Approx. price	WER (clean English)	Streaming	Diarization	Lan
Google Chirp 3	~$0.016/min	~6–9%	Yes	Yes	125
Azure Speech	~$0.017/min (real-time)	~6–9%	Yes	Yes	100
Amazon Transcribe	~$0.024/min	~7–10%	Yes	Yes	100
ElevenLabs Scribe v2	~$0.0067/min	~4–6%	No	Yes	99
Mistral Voxtral	~$0.003–0.006/min	~6–8%	No	No	8

5.2 Open weights

Model	Params	License	English WER	Languages	Speed	Notes
Whisper large-v3	1.5B	MIT	~7–9%	99	Medium	The open standard; robust
Whisper large-v3-turbo	809M	MIT	~8–10%	99	~8× faster	Best trade-off
distil-whisper large-v3.5	756M	MIT	~9–11%	en	6× faster	English
NVIDIA Parakeet-TDT-0.6B v3	0.6B	CC-BY-4.0	~5–6%	25 (European)	Very fast	Tops the Open ASR leaderboard
NVIDIA Canary-Qwen-2.5B	2.5B	CC-BY-4.0	~5.6%	en	Medium	Accuracy plus LLM
NVIDIA Canary-1B-v2	1B	CC-BY-4.0	~6–7%	25	Fast	Also does speech translation
Kyutai STT	1B / 2.6B	CC-BY-4.0	~6–8%	en, fr	Streaming	Real-time
Moonshine	27–61M	MIT	~9–12%	en	Very fast	Edge devices
Vosk	50–500M	Apache 2.0	~10–15%	20+ (incl. fa)	CPU	Small, offline

Model	Params	License	English WER	Languages	Speed	Notes
MMS / Wav2Vec2	300M	MIT	Variable	1100+ (incl. fa)	Medium	Unmatched language coverage
SenseVoice	~234M	Apache 2.0	~7–9%	zh, en, ja, ko	Very fast	Emotion and audio events
Qwen3-ASR	1.7B–8B	Apache 2.0	~5–7%	52	Medium	High accuracy, LLM-based
Step-Audio 2 mini ASR	~8B	Apache 2.0	~5–7%	30+	Slow	Part of an Omni model

Bottom line on ASR. Best accuracy: Nova-3, Universal-3, Scribe v2, Canary-Qwen-2.5B. Cheapest cloud: Soniox, Gladia, Speechmatics batch. Best open self-host: Whisper large-v3-turbo for breadth, Parakeet for speed on European languages, Canary for accuracy. Persian: Whisper large-v3 (WER ~10–15% depending on domain), Azure fa-IR, Google fa-IR, Speechmatics, Vosk fa for lightweight use, and NVIDIA NeMo Persian if you have training data.

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6. Voice conversion and cloning

Model / service	Type	License / price	Timbre similarity	Real-time	Notes
RVC	Open weights	MIT	High (with 10–30 min of data)	Yes (~90 ms)	The community workhorse; easy to train
so-vits-svc	Open weights	AGPL	High	Near	Popular for singing
OpenVoice v2 (MyShell)	Open weights	MIT	Medium	Yes	Tone-color cloning, lightweight
Seed-VC	Open weights	Apache 2.0	High	Yes	Zero-shot, speech and singing
GPT-SoVITS	Open weights	MIT	High	No	Strong quality from ~1 min of data
Chatterbox VC	Open weights	MIT	High	No	Same family as the TTS model
ElevenLabs Voice Changer	Cloud	Subscription	High	Yes	Simple, good quality

Model / service	Type	License / price	Timbre similarity	Real-time	Notes
Respeecher	Cloud / studio	Enterprise (expensive)	Very high	No	Film and TV, with voice rights
Altered Studio	Cloud	Subscription	High	Yes	Dubbing and games
Voice.ai / Kits VC	Cloud / desktop	Freemium	Medium–high	Yes	Consumer

What matters here. Cloning is a legal risk, not just a technical one. Under the EU AI Act and several US state laws, disclosing synthetic voice and obtaining consent from the voice owner is mandatory in many contexts. Take watermarking (e.g., Perth in Chatterbox) and consent logging seriously. RVC is the best cost/quality route to a custom voice; Seed-VC is the zero-shot option. For singing, RVC and so-vits-svc remain the community standard, while full-song generation is easier through Suno or ACE-Step.

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7. Music generation

7.1 Commercial and consumer

Service	Version	What it does	Approx. price	Output ownership
Suno	v5	Full songs with vocals, many styles, section editing	Free (50 credits/day, non-commercial); Pro ~$10/mo; Premier ~$30/mo	Commercial on paid plans (with terms)
Udio	v2/v3	High audio quality, style control	~$10–30/mo	Commercial on paid plans; after deals with publishers
Riffusion	FUZZ	Free/freemium, fast generation	Free + paid tiers	Varies
Stable Audio 2.5 (Stability)	2.5	Up to ~3-minute generations, loops, audio-to-audio	API credits	Commercial under license
Google Lyria 2/3	Vertex/Gemini	High-fidelity music, pro tools	Per minute/token	Commercial
MiniMax Music 2	—	Full songs, cheap	Cheap API	Commercial
Mureka	V8	Full songs with vocals	Subscription	Commercial
ElevenLabs Music	—	Licensed-data music, commercial focus	Subscription	Commercial (fully licensed claim)

Service	Version	What it does	Approx. price	Output ownership
Adobe Firefly Soundtrack	—	Copyright-safe	Creative Cloud credits	Commercial
Soundraw / Beatoven / AIVA / Boomy	—	Royalty-free music for video	$10–30/mo	Commercial

7.2 Open weights

Model	Params	License	Length	Vocals	Notes
ACE-Step	3.5B	Apache 2.0	up to ~4 min	Yes (19 languages)	The most complete open option for full songs
Stable Audio Open 2.0	~1B	Stability Community (commercial under $1M revenue)	~47 s	No	Loops and effects, high quality
MusicGen (Meta)	300M–3.3B	Code MIT / weights CC-BY-NC	30 s	No	Research reference; melody/style
YuE	~7B	Code Apache / weights CC-BY-NC	several minutes	Yes	Full songs with vocals
DiffRhythm	~1B	Apache 2.0	up to ~4 min	Yes	Very fast (seconds)
Tango 2	~1B	CC-BY-NC	30 s	No	Strong SFX/music quality
AudioLDM 2	~1B	CC-BY-NC	10 s	No	General audio
MAGNeT	300M–1.5B	CC-BY-NC	30 s	No	Fast, non-autoregressive

Bottom line on music. For release-ready songs, Suno and Udio. For legally safe commercial background music, Firefly, Stable, or ElevenLabs Music. For local deployment and control, ACE-Step (best open) and DiffRhythm (fastest).

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8. Sound effects and Foley

Model / service	Input	License / price	Notes
ElevenLabs SFX v2	text	Subscription/credits	High quality, sound library, video sync
Stable Audio Open / 2.5 API	text	Community / credits	Effects and loops
Meta AudioGen	text	CC-BY-NC	Research reference
TangoFlux	text	Apache 2.0	515M, fast, good quality
Woosh	text	Apache 2.0	Lightweight, effects-focused
MMAudio	video (+text)	MIT	Automatic Foley for video
HunyuanVideo-Foley (Tencent)	video	Open (research)	Tight picture sync
AudioX	video/text	Open	Audio from video
Kling / Runway / Pika Foley	video	Subscription	Inside video tools
Mirelo SFX	video	API	Professional Foley

What matters here. For video, the metric is temporal sync, not just audio quality. Video-to-audio models (MMAudio, HunyuanVideo-Foley) beat text-to-audio on that metric.

9. Singing synthesis

Option	Type	Price / license	Quality	Notes
Synthesizer V Studio (Dreamtonics)	Commercial software	~$89 + voicebanks $50–100	Best in class	Full phonetic and vibrato control
Vocaloid 6 / CeVIO / Piapro	Commercial	$100–300	High	Large legacy ecosystem
Suno / Udio	Cloud	Subscription	High (generated vocals)	Limited fine control
Kits.AI / Voicify / Jammable	Cloud	Subscription	Medium–high	Voice cloning on singing
DiffSinger	Open	Apache 2.0	Medium–high	Needs data/training
ACE-Step / DiffRhythm / YuE	Open	Apache/NC	Medium	Full vocals inside songs
RVC / so-vits-svc / Seed-VC	Open	MIT/AGPL	High (conversion)	Convert an existing vocal to a target voice

What matters here. Singing from scratch (lyrics + notes) and voice conversion (a real singer → another timbre) are different problems. For studio-grade results Synthesizer V still wins; for speed and cost, Suno.

10. Audio understanding (Audio LLMs)

Model	License	Capability	Notes
GPT-4o / GPT-5 audio (OpenAI)	API	Listening, reasoning, spoken replies, Realtime	The most expensive and the strongest in conversation
Gemini 2.5 / 3 (Google)	API	Audio + video + text, long context	Deep Gemini integration
Qwen3-Omni (Alibaba)	Apache 2.0	Audio/image/text, 30B-A3B MoE	The best open Omni model
Qwen2.5-Omni	Apache 2.0	Real-time conversation	Lighter
Kimi-Audio (Moonshot)	MIT-ish	Understanding + generation, 13M hours of data	Strong benchmark scores
Step-Audio 2 mini	Apache 2.0	ASR + TTS + audio reasoning	Strong in Chinese and English
Moshi (Kyutai)	CC-BY 4.0	Full-duplex conversation (~200 ms)	Feels genuinely conversational
SALMONN / Audio Flamingo 3	Research	General audio and music understanding	Non-commercial
GLM-4-Voice / MiniCPM-o	Apache 2.0	Lightweight conversation	Edge and mobile

Where these get used: voice assistants, call analytics, factory monitoring, music description, accessibility. The benchmarks that matter are MMAU, AudioBench, and LongAudioBench.

11. Processing, separation, and restoration

Tool / model	Task	License	Notes
Demucs v4 (htdemucs)	4/6-stem separation	MIT	Open standard; good quality
BS-RoFormer / Mel-Band RoFormer	Stem separation	MIT-ish	Best separation quality
MDX23C / UVR5	Separation + GUI	GPL/MIT	Community favorite
LavaSR	Bandwidth/quality extension	Apache 2.0	Audio reconstruction
AudioSR	Audio super-resolution	MIT	44.1/48 kHz
Apollo	Restoration of degraded audio	Open	Loss restoration
DeepFilterNet3	Real-time noise suppression	MIT	Light, built for calls
RNNoise	Noise suppression	BSD	Ultra-light
Resemble Enhance	Speech enhancement	MIT/Apache	Denoise + enhance
Adobe Podcast Enhance	Speech enhancement	Free (web)	Good quality, limited
Auphonic / iZotope RX	Mastering/restoration	Subscription/license	Professional standard
Krisp	Noise/echo removal	Subscription	SDK for calls

12. The cost question

12.1 Cost per unit of work (approximate, November 2026)

Work	Unit	Cloud range	Equivalent per hour
Cheap TTS (Polly/Azure/OpenAI tts-1)	1M chars	$4–16	**$0.40–1.50**
Mid-tier TTS (Cartesia/Deepgram/Fish)	1M chars	$15–30	**$1.40–2.70**
Premium TTS (ElevenLabs/Hume)	1M chars	$60–240	**$5–22**
Cheap ASR (Soniox/Gladia/Speechmatics batch)	minute	$0.001–0.002	**$0.06–0.12**
Mid-tier ASR (Deepgram/AssemblyAI)	minute	$0.0025–0.005	**$0.15–0.30**
Premium ASR (Google/Azure/Rev)	minute	$0.016–0.02	**$0.96–1.20**
Music (Suno/Udio)	track	—	$10–30/mo subscription
Music API (Stable/Lyria)	minute	$0.02–0.1	**$1.20–6**
Sound effects	generation	$0.01–0.1	varies

One hour of speech is about 90,000 characters; one hour of audio is 60 minutes. Do not mix the two.

12.2 Self-hosting TCO

Item	Approximate cost
Cloud GPU (L4/A10G)	$0.5–1.2 per hour
Cloud GPU (A100/H100)	$2–4 per hour
RTX 4090 (rented)	$0.3–0.7 per hour
Dedicated GPU server	$300–1,500 per month
MLOps/maintenance	15–30% of infrastructure

The key sentence. Whisper large-v3-turbo on a single L4 transcribes at roughly 30–60× real time — so one GPU-hour equals 30–60 hours of audio, or about **$0.01–0.04 per audio hour**. At scale (above roughly 500 hours per month) that undercuts any cheap API. What it does not include is engineering time and operational risk.

Break-even. If you produce or transcribe more than 300–500 hours of audio per month and have engineering capacity, self-hosting deserves serious consideration. Below that, APIs are cheaper and far less painful.

12.3 Hidden costs

- **Egress and bandwidth** on bulk audio delivery.
- **Retries and quality:** TTS error rates (mispronunciation, truncation) cause regeneration and double billing.
- **Version lock:** cloud models change without notice; yesterday's output is not today's. Pin versions or cache outputs.
- **Data governance:** voice is biometric data. In some sectors (health, banking), sending it to the cloud is prohibited, which forces self-hosting.

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13. Licensing: a decision table

License	Commercial use	Modification	Attribution	Examples
Apache 2.0	Yes	Yes	Yes (NOTICE)	Kokoro, CosyVoice, Dia, Orpheus, Qwen3-Omni
MIT	Yes	Yes	Yes	Whisper, Piper, Chatterbox, Demucs, RVC
BSD	Yes	Yes	Yes	RNNoise
CC-BY 4.0	Yes	Yes	Yes	Moshi, Parakeet/Canary
CC-BY-NC	No	Yes	Yes	MusicGen weights, F5-TTS weights, YuE weights
Stability Community	Yes below $1M revenue	Yes	Yes	Stable Audio Open
Llama/Community proprietary	Conditional	Conditional	—	Some Meta/Microsoft models
CPML (Coqui)	No	Yes	—	XTTS-v2
AGPL	Yes (with service source obligation)	Yes	Yes	so-vits-svc
Proprietary cloud	Per contract	No	—	ElevenLabs, Suno, Deepgram

The golden rule: the license on the repository is not the license on the weights. F5-TTS has MIT code and CC-BY-NC weights; Fish Speech has open code and non-commercial weights. Read the `LICENSE` file attached to the weights on Hugging Face.

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14. Choosing by scenario

Scenario	First choice	Alternative	Why
Phone agent	Cartesia Sonic-3 + Deepgram Nova-3	ElevenLabs Flash + AssemblyAI	Sub-300 ms latency, predictable cost
Persian audiobook	Azure Neural (fa-IR) + editing	ElevenLabs (test first)	Official Persian coverage and pronunciation
English podcast/video	ElevenLabs v3	PlayHT, Hume	Expression and cloning
Film/TV dubbing	Respeecher / ElevenLabs + processing	Synthesizer V for singing	Quality and voice rights
Customer calls (ASR + analytics)	AssemblyAI / Deepgram	Speechmatics	Diarization plus conversation analytics
Bulk cheap transcription	Whisper large-v3-turbo (self-host)	Soniox API	Lowest cost at scale
Persian subtitles	Whisper + human review	Azure fa-IR	Acceptable accuracy, low cost
Commercial background music	Adobe Firefly / Stable Audio	ACE-Step (self-host)	Legal safety
Full songs with vocals	Suno v5 / Udio	ACE-Step / YuE	Quality and speed
Game sound effects	ElevenLabs SFX / Stable Audio	TangoFlux, Woosh	Variety and control

Scenario	First choice	Alternative	Why
Automatic video Foley	MMAudio / HunyuanVideo-Foley	Kling Foley	Picture sync
Call noise suppression	DeepFilterNet3 / Krisp	RNNoise	Real-time and light
Stem separation for remixes	BS-RoFormer / Demucs	UVR5	Separation quality
Research and fine-tuning	Whisper + Qwen3-Omni + ACE-Step	Kokoro, Parakeet	Open and modifiable

15. The Persian situation

Persian ASR. Whisper large-v3 works on Persian (WER around 10–15% on clean speech, higher in casual or noisy audio). Common failure modes: numbers, the zero-width non-joiner, Arabic versus Persian letter forms, and proper nouns. Azure fa-IR and Google fa-IR have official support and are solid on formal or news-style speech. Speechmatics supports Persian; Vosk fa is light but inaccurate; MMS covers it with lower quality. For high accuracy on conversational Persian, fine-tuning Whisper on labeled in-domain data is the most reliable path.

Persian TTS. Azure and Google are the stable official options. ElevenLabs covers Persian in its multilingual models, but pronunciation quality needs testing on your own vocabulary — numbers, idioms, names. On the open side: Piper fa_IR (fast, light, robotic) and community VITS/Coqui models trained on ManaTTS and Common Voice fa. The quality ceiling depends on data and training time.

The real gap. No open multilingual model at the current state of the art covers Persian at English-level quality. For a Persian product, a fine-tuned open model plus a pronunciation dictionary is

more realistic than expecting a general model to handle it out of the box.

Engineering notes for Persian: text normalization (digits, half-spaces, alef variants), a pronunciation lexicon for proper nouns, and evaluation on an internal test set rather than public MOS numbers.

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16. Benchmarks worth looking at

Area	Benchmark / leaderboard	Caveat
TTS	**TTS Arena (Hugging Face)**, **Artificial Analysis Speech Arena**	Human votes; correlates well with perception
ASR	**Open ASR Leaderboard (HF)**	English WER; not valid for Persian
Multilingual ASR	**FLEURS**, **Common Voice**	Language coverage
Audio understanding	**MMAU**, **AudioBench**, **LongAudioBench**	Reasoning and long-form audio
Separation	**MUSDB18 / SDR**	Stem quality
Music	Human evaluation + FAD	Automated metrics are not enough

A warning. Public benchmarks are not your domain. Always build a small test set from your own data (50–200 samples) and compare models on it. That is the only metric you can actually trust.

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17. Risks and legal considerations

1. **Voice cloning** requires explicit consent, consent logging, and disclosure that the audio is synthetic. The EU AI Act and several US states (Tennessee, California) impose transparency requirements.
2. **Music copyright.** Suno and Udio reached agreements with publishers in 2025, but open models trained on protected data still carry risk. For commercial products, choose models with transparent data licensing.
3. **Biometric data.** Voice is sensitive personal data under GDPR and CCPA; storing or transferring it requires a DPIA.
4. **Vendor lock-in.** Dependence on one API means exposure to price and version changes. Keep an abstraction layer and a swap path.
5. **Safety and hallucination.** Audio models can generate harmful speech; output filtering and logging are necessary.
6. **Low-resource languages.** Quality drops sharply versus English. Manage expectations accordingly.

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18. A practical evaluation checklist

□ 50–200 real samples from your own data (including noise, accents, terminology).

□ Numeric success criteria: target WER, MOS, TTFA, pronunciation error rate.

□ Blind A/B testing between the three finalists.

□ Cost per unit of work including retry rates.

□ Weight license review for commercial use.

□ Stability test: same input, ten runs, measure variance.

□ Streaming/latency test on real networks, not just benchmarks.

□ An exit plan: export format and model replacement path.

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19. Sources

Leaderboards and benchmarks:

- Artificial Analysis — Text to Speech Leaderboard
- Hugging Face — Open ASR Leaderboard
- Hugging Face — TTS Arena

Pricing and comparison (for verification):

- Deepgram Pricing
- ElevenLabs Pricing
- OpenAI API Pricing
- AssemblyAI Pricing
- Google Cloud TTS Pricing
- Azure Speech Pricing
- AWS Polly Pricing
- Suno Pricing
- Future AGI — TTS comparison 2026
- Future AGI — STT comparison 2026

Open models (selection):

- [Kokoro-82M](https://huggingface.co/hexgrad/Kokoro-82M)
- Chatterbox
- CosyVoice

- F5-TTS
- ACE-Step
- Demucs
- Qwen3-Omni
- Whisper

Appendix: ten quick picks

If you need…	Pick
Cheapest good TTS	Kokoro-82M (self-host) or OpenAI tts-1
Best commercial TTS	ElevenLabs v3
Best commercial ASR	Deepgram Nova-3 / AssemblyAI Universal-3
Best open ASR	Whisper large-v3-turbo or Parakeet-TDT
Best full songs	Suno v5
Best open music	ACE-Step
Best text-to-SFX	ElevenLabs SFX v2
Best video Foley	MMAudio / HunyuanVideo-Foley
Best open audio understanding	Qwen3-Omni
Best stem separation	BS-RoFormer